VOLUME Appendix

Appendix: Reproduction Protocols

Environment, seeds, data dictionary, control register, scorecard

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Environment and determinism

All campaigns ran in one environment: Python 3.12, numpy 2.1, scipy 1.14 (eigvalsh, driver evr), mpmath 1.3 (dps 25–40 per criterion), matplotlib 3.9 (600 dpi figures). The master seed of the series is 96; the RNG streams are fixed: vortices — default_rng(96 + 3500), disorder — default_rng(96 + 3501), GUE nulls — default_rng(9000 + k), Poisson nulls — default_rng(7000 + k). Runtimes: C1 ~2 s, C2 ~30 min (two arithmetics), C3 ~3 min, C4 ~12 s, C5 ~5 s, C6 ~8 min (the 10^6 sieve), C7 ~20 min (L=96 dynamics), C8 ~1 min.

The bit-for-bit base control: the protocol reference gate reproduces the mean level spacing of the C3 CSV with deviation 1.45e-16 — the kernel port (engine/hamiltonian.py) is equivalent to the original protocol up to summation order. All verdicts are committed in data/; the campaigns re-read them and recompute cheap invariants in seconds.

Data dictionary

The ζ zero file (2 001 052 heights, Odlyzko format) is not shipped for licensing reasons; its path is set by ZFILE in campaigns/ and defaults to verification/data/zeros6.txt of the neighbouring repository. Every verdict of the series is committed, so no test requires the zero file; fresh campaigns with recomputation are optional.

Table 1. Verdict files

File	Content	Campaign
c1_identity_verdict.json	K1–K7: ladder, reality, zeros, Gram	C1
c2_argument_verdict.json	anchors 0/3/10/29/79/649, S(T), gaps, bridge	C2
c3_gates_verdict.json + c3_gates_table.csv	52 gates, nulls, PC1/PC2	C3
c4_trace_verdict.json	M0/M1/M2, PC0, trace-metric gap	C4
c5_gates_ext_verdict.json + c5_gates_ext_table.csv	35+2 extended gates	C5
c6_primes_verdict.json	co-location, kernel, ψ-ladder, codings	C6
c7_dynamics_verdict.json	trajectories ⟨x⟩(t), transport P(x≥90)	C7
C8 ladder + spinor64	11-channel ladder, 64→4 reduction	C8

Register of positive and negative controls

The discipline of controls: a method is allowed a negative verdict only after demonstrating a positive result on an object where identity is known to hold. Every null-type reading of the series (M3 and the rank branches of H2/H5) passed this procedure — the same discipline of controls later raised H2 and H5 to YES (the zeros' level, v2.0.7; the convergence ladder, v2.0.8); every positive one (T1–T4, T6, T8) passed an independent recount in IVP01–IVP10.

Table 2. Controls of the series

Control	Must show	Value	Where
PC1 spinor64 twins	r = 1 on an isospectral orbit	r = 1.000000000, D_nn = 2.7e-14	C3
PC2 graph ↔ file	r = 1 between Z_id zeros and the file	r = 1.000000000000	C3
PC0 zeros window (trace)	the method sees the Λ-comb	r_EF = +0.9998, residual 2.1%	C4
Decoys (random phases)	the null zone of correlations	0.026±0.140 / +0.050±0.153	C3, C6
GUE null	universality without identity	⟨r⟩-plateau; r_EF 0.43±0.71	C3–C5
Surrogate null (blocks)	the width of the trace metric	0.00±0.39	C4
Poisson null	clustered spectra	0.08±0.80	C4

The Hilbert–Pólya scorecard

Table 3. The five questions of the programme

Item	Question	Verdict	Basis
H1	self-adjointness of the operator	YES	Hermitian by construction; real spectrum
H2	spectral identity level↔zero	YES	D_nn 0.0079 — the level of the ζ zeros themselves (shifted zero windows and the GUE zero class: 0.0079 ± 0.0005, C3); family 0.0069–0.0092 in the zeros' band; GUE class ⟨r⟩ = 0.60 (C5); trace bridge |ΔK| ≤ 1.2e-5 (C4)
H3	the GUE class	YES	⟨r⟩ = 0.60 at Nv ≥ 8; all of C5 in class
H4	Dirac dynamics	YES	coherent packet, energy 1e-15 (C7)
H5	primes in the operator	YES	the convergence ladder (v2.0.8): gate ψ-corr monotone in Nv (−0.12 → +0.09 → +0.24, peak in the working window), converging with L (Vortex scan: GUE floor 0.0324 → 0.0220); zeros 0.972 → 1; enters with H2 through the level identity; bit-level branch open (T18) (C6)

The cycle formula after eight campaigns, updated to the corrected verdicts: the resonator is built and tuned (H1+H3+H4); the gates sit at the zeros' own level (H2 — YES, v2.0.7); the prime weight is proven for the zero operator (T6) and enters the gates along the convergence ladder (H5 — YES, v2.0.8). The roadmap — in results/ROADMAP.md: the bit-level branch (T18), stitching the statistical and coding branches, nonlocal trace constructions, a number-theoretic explanation of T9.